15 May 2024

Configuration File Manual

I/O Pin Mapping and Driver Configuration

I/O Module	IO316-100k
Rear Plug-in Module	IO316-100k-21
Minimum required I/O Blockset	9.8.0.1
Reference	speedgoat_IO316_100k_21_CI_02826_v2

real-time simulation and testing

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Document Version History

Rev.	Description	Date
1.0	Initial version of Configuration File Manual	15 May 2024

1 Introduction

This manual supplements the User Manual you received with your real-time target machine.

Your real-time target machine is equipped with at least one configurable I/O module for which a configuration file and Simulink Real-Time ™ driver blockset have been provided. Refer to your real-time target machine user manual for more information.

This document explains how to install, configure, and use the specific configuration file and Simulink Real-Time driver blockset you ordered upon purchase of your real-time target machine. Note that additional configuration files can also be purchased at a later date.

2 Implemented Functionality

This configuration file and Simulink Real-Time driver blockset implements the following functionality:

FPGA Code Module	Transceiver Type	No.of Modules/Channels	Version
CAP	TTL	36	1.15
DIO	TTL	120	1.3
INTA	TTL	1	1.5
PWM	TTL	5	5.12
QAE	TTL	2	3.4
SPI	TTL	2	2.4

Table 1: Implemented Functionality

3 Software Installation

Install the configuration file after installing the MathWorks tool chain and the Speedgoat I/O Blockset. The steps are as follows:

1. Download the configuration file archive, listed in the downloads/software section of the Speedgoat Customer Portal. If the archive is not available, please contact Speedgoat support.
2. The archive contains a test model, library, documentation and the configuration file (*.mat).
3. The content of this archive can be extracted to your current MATLAB workspace. The configuration file must be part of the MATLAB path. For your convenience, you can copy the file to the following directory, which is part of the MATLAB path. To obtain the predefined directory, in MATLAB type:
   `» fullfile(speedgoatroot,'sg_bitstream')`
4. After copying the MAT-file, type:
   `» rehash toolbox`

4 I/O Pin mapping

The I/O pin mapping for this configuration file implementation is shown below. This I/O pin mapping is specific to the configurable I/O module for which it has been designed and shows how to connect the pins to your hardware under test.

> **Note:** Speedgoat delivers real-time target machines together with terminal boards (refer to your target machine user manual). The terminal board for the configurable I/O module must therefore be wired as described in section 7.2 of this document in order to execute the test model for this specific implementation.

4.1 Front I/O

Pin	Digital I/O	Code Module	Functionality	Direction	Transceiver
	Channel number				
1	1	1	SPI - CLK	IN/OUT	TTL
2	2	1	SPI - CS1	IN/OUT	TTL
3	3	1	SPI - SDO	OUT	TTL
4	4	1	SPI - SDI	IN	TTL
5	5	2	SPI - CLK	IN/OUT	TTL
6	6	2	SPI - CS1	IN/OUT	TTL
7	7	2	SPI - SDO	OUT	TTL
8	8	2	SPI - SDI	IN	TTL
9			Ground		
10	9	1	PWM - A	OUT	TTL
11	10	1	PWM - B	OUT	TTL
12	11	1	PWM - Trigger	OUT	TTL
13	12	2	PWM - A	OUT	TTL
14	13	2	PWM - B	OUT	TTL
15	14	2	PWM - Trigger	OUT	TTL
16	15	3	PWM - A	OUT	TTL
17	16	3	PWM - B	OUT	TTL
18	17	3	PWM - Trigger	OUT	TTL
19	18	4	PWM - A	OUT	TTL
20	19	4	PWM - B	OUT	TTL
21	20	4	PWM - Trigger	OUT	TTL
22	21	5	PWM - A	OUT	TTL
23	22	5	PWM - B	OUT	TTL
24	23	5	PWM - Trigger	OUT	TTL
25	24	1	CAP	IN	TTL
26			Ground		
27	25	1	CAP - Trigger	IN	TTL
28	26	2	CAP	IN	TTL
29	27	2	CAP - Trigger	IN	TTL
30	28	3	CAP	IN	TTL
31	29	3	CAP - Trigger	IN	TTL
32	30	4	CAP	IN	TTL
33	31	4	CAP - Trigger	IN	TTL
34	32	5	CAP	IN	TTL

Pin	Digital I/O	Code Module	Functionality	Direction	Transceiver	Port	Pull Resistors
	Channel number						
35	33	5	CAP - Trigger	IN	TTL	1	pull-up 3.3 VDC weak pull-up 5.0 VDC pull-down, or floating
36	34	6	CAP	IN	TTL		
37	35	6	CAP - Trigger	IN	TTL		
38	36	7	CAP	IN	TTL		
39	37	7	CAP - Trigger	IN	TTL		
40	38	8	CAP	IN	TTL		
41	39	8	CAP - Trigger	IN	TTL		
42	40	9	CAP	IN	TTL		
43			Ground				
44	41	9	CAP - Trigger	IN	TTL	2	pull-up 3.3 VDC weak pull-up 5.0 VDC pull-down, or floating
45	42	10	CAP	IN	TTL		
46	43	10	CAP - Trigger	IN	TTL		
47	44	11	CAP	IN	TTL		
48	45	11	CAP - Trigger	IN	TTL		
49	46	12	CAP	IN	TTL		
50	47	12	CAP - Trigger	IN	TTL		
51	48	13	CAP	IN	TTL		
52	49	13	CAP - Trigger	IN	TTL	3	pull-up 3.3 VDC weak pull-up 5.0 VDC pull-down, or floating
53	50	14	CAP	IN	TTL		
54	51	14	CAP - Trigger	IN	TTL		
55	52	15	CAP	IN	TTL		
56	53	15	CAP - Trigger	IN	TTL		
57	54	16	CAP	IN	TTL		
58	55	16	CAP - Trigger	IN	TTL		
59	56	17	CAP	IN	TTL		
60			Ground				
61	57	17	CAP - Trigger	IN	TTL	4	pull-up 3.3 VDC weak pull-up 5.0 VDC pull-down, or floating
62	58	18	CAP	IN	TTL		
63	59	18	CAP - Trigger	IN	TTL		
64	60	19	CAP	IN	TTL		
65	61	19	CAP - Trigger	IN	TTL		
66	62	20	CAP	IN	TTL		
67	63	20	CAP - Trigger	IN	TTL		
68	64	21	CAP	IN	TTL		

Figure 1: IO316-100k front pinout

4.2 Rear I/O

Pin	Digital I/O	Code Module	Functionality	Direction	Transceiver	Port	Pull Resistors
	Channel number						
1	65	21	CAP - Trigger	IN	TTL	1	pull-up 3.3 VDC weak pull-up 5.0 VDC pull-down, or floating
2	66	22	CAP		TTL		
3	67	22	CAP - Trigger		TTL		
4	68	23	CAP		TTL		
5	69	23	CAP - Trigger		TTL		
6	70	24	CAP		TTL		
7	71	24	CAP - Trigger		TTL		
8	72	25	CAP		TTL		
9			Ground				
10	73	25	CAP - Trigger	IN	TTL	2	pull-up 3.3 VDC weak pull-up 5.0 VDC pull-down, or floating
11	74	26	CAP		TTL		
12	75	26	CAP - Trigger		TTL		
13	76	27	CAP		TTL		
14	77	27	CAP - Trigger		TTL		
15	78	28	CAP		TTL		
16	79	28	CAP - Trigger		TTL		
17	80	29	CAP		TTL		
18	81	29	CAP - Trigger	IN	TTL	3	pull-up 3.3 VDC weak pull-up 5.0 VDC pull-down, or floating
19	82	30	CAP		TTL		
20	83	30	CAP - Trigger		TTL		
21	84	31	CAP		TTL		
22	85	31	CAP - Trigger		TTL		
23	86	32	CAP		TTL		
24	87	32	CAP - Trigger		TTL		
25	88	33	CAP		TTL		
26			Ground				
27	89	33	CAP - Trigger	IN	TTL	4	pull-up 3.3 VDC weak pull-up 5.0 VDC pull-down, or floating
28	90	34	CAP		TTL		
29	91	34	CAP - Trigger		TTL		
30	92	35	CAP		TTL		
31	93	35	CAP - Trigger		TTL		
32	94	36	CAP		TTL		
33	95	36	CAP - Trigger		TTL		
34	96	1	Interrupt input		TTL		

Pin	Digital I/O	Code Module	Functionality	Direction	Transceiver	Port	Pull Resistors
	Channel number						
35	97	1	QAE - A	OUT	TTL	5	pull-up 3.3 VDC weak pull-up 5.0 VDC pull-down, or floating
36	98	1	QAE - B		TTL		
37	99	1	QAE - C/Index		TTL		
38	100	2	QAE - A		TTL		
39	101	2	QAE - B		TTL		
40	102	2	QAE - C/Index		TTL		
41	103	103	DIO		TTL		
42	104	104	DIO		TTL		
43			Ground				
44	105	105	DIO	IN/OUT	TTL	6	pull-up 3.3 VDC weak pull-up 5.0 VDC pull-down, or floating
45	106	106	DIO		TTL		
46	107	107	DIO		TTL		
47	108	108	DIO		TTL		
48	109	109	DIO		TTL		
49	110	110	DIO		TTL		
50	111	111	DIO		TTL		
51	112	112	DIO		TTL		
52	113	113	DIO	IN/OUT	TTL	7	pull-up 3.3 VDC weak pull-up 5.0 VDC pull-down, or floating
53	114	114	DIO		TTL		
54	115	115	DIO		TTL		
55	116	116	DIO		TTL		
56	117	117	DIO		TTL		
57	118	118	DIO		TTL		
58	119	119	DIO		TTL		
59	120	120	DIO		TTL		
60			Ground				
61			Reserved		TTL	-	pull down
62			Reserved		TTL		
63			Reserved		TTL		
64			Reserved		TTL		
65			Reserved		TTL		
66			Reserved		TTL		
67			+5V/24mA supply				
68			+5V/24mA supply				

Figure 2: IO316-100k-21 rear plug-in pinout

5 Simulink Driver Library

To open the library blocks used for the configuration file, type:
» `speedgoat_lib_IO316_100k_21_CI_02826_v2`

at the MATLAB command line prompt. The blockset appears as follows:

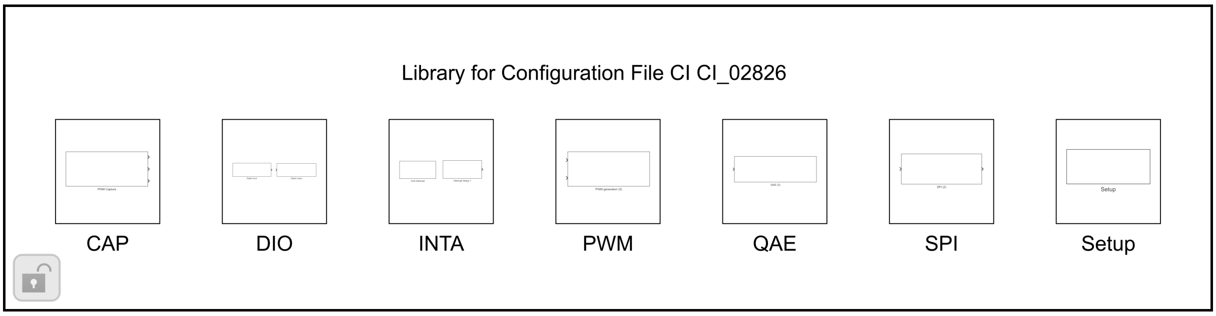

Figure 3: Library for Custom Implementation CI_02826

The configuration file library contains a subset of the Speedgoat I/O Blockset driver blocks implemented in this specific bitstream.

The Speedgoat I/O Blockset may contain different versions of the driver blocks which are not compatible with your configuration file. We therefore highly recommend that you use this custom library whenever you start to build a new Simulink model.

Alternatively, ensure you select the right version of the respective driver blocks. The version of your implemented Code Module functionality is listed in Section 2, where the major version number signifies the driver block version.

6 Configuration File

The configuration file is specified in the setup driver block.

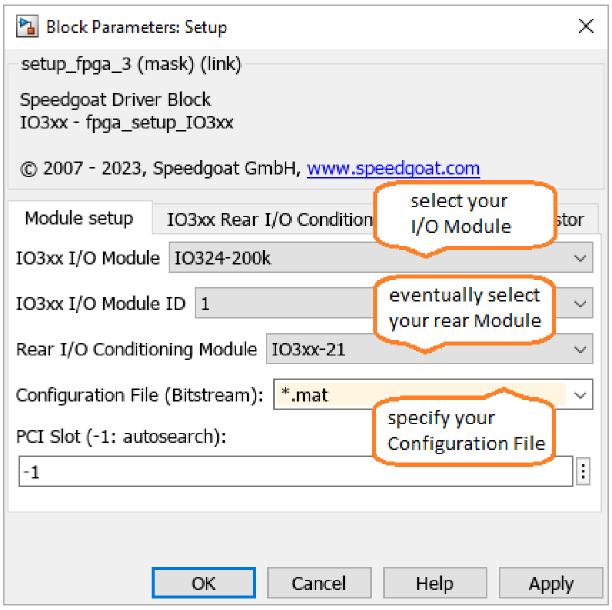

Figure 4: Setup block mask parameter

Select following parameters for your configuration file

- I/O Module: IO316-100k
- Rear I/O Conditioning Module: IO3xx-21
- Configuration File: speedgoat_IO316_100k_21_CI_02826_v2.mat

The FPGA frequency and the resulting tick time (see figure above) are used for time measurements and configuration in several Code Modules (like PWM, CAP) and for defining transmission frequencies for protocols (such as SPI, I2C).

7 Test Model

7.1 Test Model Description

A dedicated test model is included to test the custom set of Code Modules.

Note that this test model only tests I/O channels for which the loop-back test method is possible. The terminal board provided must be wired as indicated in chapter 7.2 of this manual.

To open the test model type:
» `speedgoat_IO316_100k_21_CI_02826_v2`

at the MATLAB command line prompt. The model appears as follows:

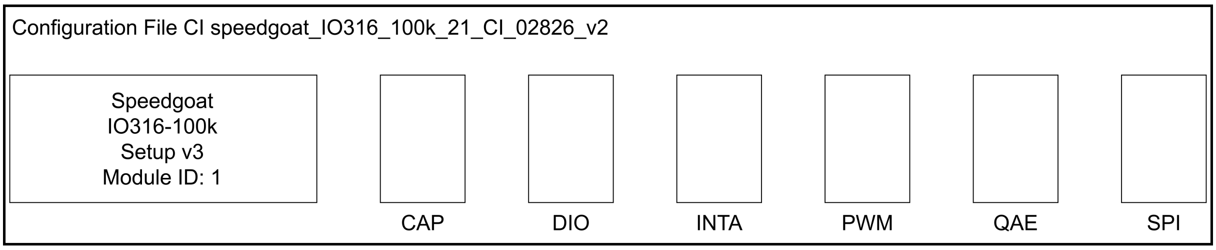

Figure 5: Test model speedgoat_IO316_100k_21_CI_02826_v2

7.2 Required Test Wiring of the Terminal Board

Front I/O:

From Pin	To Pin	Tested Functionality
10	25	PWM - A channel 1 to CAP channel 1
12	27	PWM - Trigger channel 1 to CAP - Trigger channel 1
11	28	PWM - B channel 1 to CAP channel 2
3	4	SPI - SDO channel 1 to SPI - SDI channel 1
5	6	SPI - CLK channel 2 to SPI - CS channel 2

Table 2: Terminal Board wiring IO316-100k Front I/O

Rear Plugin I/O

From Pin	To Pin	Tested Functionality
41	34	DIO channel 103 to INTA channel 1
35	44	QAE - A channel 1 to DIO channel 105
36	45	QAE - B channel 1 to DIO channel 106
37	46	QAE - C_Index channel 1 to DIO channel 107
42	47	DIO channel 104 to DIO channel 108

Table 3: Terminal Board wiring IO316-100k-21 Rear I/O